

Supplement to Quaternion Nondecimated Wavelet Descriptors for Multiclass Breast Histology Classification

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Abstract

This supplementary material provides the mathematical and computational details supporting the main paper. It records the quaternion algebra, RGB-to-quaternion image coding, the two-sided nondecimated quaternion wavelet filtering convention, the synthesis formula, and the coefficient-level interpretation of the QNDWT2D representation. It also gives explicit definitions of the retained feature classes and individual predictors used in the radial-SVM permutation-importance analysis, including energy, amplitude, phase, axis, orientation, RGB marginal, and across-level summaries.

The material is separated from the main paper to keep the main text focused on motivation, data, classification results, and histopathological interpretation, while preserving enough detail for reproducibility. In particular, the supplement clarifies which features are computed from the original RGB image and which are computed after QNDWT2D filtering, how the quaternion components are interpreted, and how redundant predictors were handled before reporting the nonredundant importance summaries.

1 Quaternion Algebra and Color Image Coding

This supplement summarizes the quaternion notation used in the paper and gives the operator form of the nondecimated quaternion wavelet transform used in the computations. More extensive discussions of quaternion and hypercomplex wavelet transforms for color vector images can be found in Bülow [4], Bayro-Corrochano [2], Chan et al. [5], Fletcher and Sangwine [10], and Fletcher [9].

A quaternion is an element of the four-dimensional real algebra

$$q = a + bi + cj + dk, \quad (1)$$

where $a, b, c, d \in \mathbb{R}$. The imaginary units satisfy

$$i^2 = j^2 = k^2 = -1, \quad (2)$$

and

$$ij = k, \quad jk = i, \quad ki = j, \quad (3)$$

23 with anti-commutation relations

$$ji = -k, \quad kj = -i, \quad ik = -j. \quad (4)$$

24 Thus quaternion multiplication is noncommutative. This is important in transform design because
25 the order of left and right multiplication must be specified.

26 The conjugate of q is

$$\bar{q} = a - bi - cj - dk, \quad (5)$$

27 and the modulus is

$$|q| = \sqrt{q\bar{q}} = \sqrt{a^2 + b^2 + c^2 + d^2}. \quad (6)$$

28 The scalar and vector parts of q are

$$\text{Sc}(q) = a, \quad (7)$$

$$\text{Vec}(q) = bi + cj + dk. \quad (8)$$

29 A pure quaternion has zero scalar part. Thus a pure quaternion is of the form

$$q = bi + cj + dk. \quad (9)$$

30 For an RGB image, let $R(x, y)$, $G(x, y)$, and $B(x, y)$ denote the normalized red, green, and blue
31 intensities at pixel location (x, y) . We encode the image as a pure quaternion-valued field

$$A_0(x, y) = R(x, y)i + G(x, y)j + B(x, y)k. \quad (10)$$

32 The scalar part is therefore initially zero. After quaternion filtering, however, the resulting coeffi-
33 cients may be full quaternions with nonzero scalar and vector parts. This is expected and reflects
34 the coupling introduced by quaternion multiplication.

35 This representation treats each RGB pixel as one vector-valued algebraic object, rather than as
36 three unrelated scalar measurements. It is especially natural for histology images, where staining,
37 morphology, and local texture are jointly expressed through the interaction of the color channels
38 [11, 23, 14].

39 Quaternion Wavelet Filter.

40 A quaternion filter is a finite sequence of quaternion coefficients. We write a low-pass filter and
41 a high-pass filter as

$$h = \{h_0, h_1, \dots, h_{K-1}\}, \quad (11)$$

$$g = \{g_0, g_1, \dots, g_{K-1}\}, \quad (12)$$

where each coefficient is a quaternion,

$$h_r = h_{r0} + h_{r1}i + h_{r2}j + h_{r3}k, \quad (13)$$

$$g_r = g_{r0} + g_{r1}i + g_{r2}j + g_{r3}k. \quad (14)$$

At transform level j , the filters are dilated as in the nondecimated wavelet transform. If the original filter has length K , the level j filter is obtained by inserting $2^{j-1} - 1$ zero quaternion coefficients between consecutive original filter taps [17, 21].

Equivalently, the level j dilated low-pass filter $h^{(j)}$ satisfies

$$h_{r2^{j-1}}^{(j)} = h_r, \quad r = 0, \dots, K-1, \quad (15)$$

and is zero at all other positions. The same dilation rule is used for $g^{(j)}$.

The implementation used in this paper uses a length-10 quaternion filter pair. The low-pass coefficients are

$$\begin{aligned} h_0 &= -0.0115088994 - 0.0200400312i, \\ h_1 &= -0.0391302581 + 0.0224723305i + 0.0166753392k, \\ h_2 &= 0.0073656956 + 0.2076972100i + 0.0500260177k, \\ h_3 &= 0.2393851082 + 0.1226724866i + 0.0166753392k, \\ h_4 &= 0.5109951348 - 0.3328019959i - 0.0833766962k, \\ h_5 &= h_4, \\ h_6 &= h_3, \\ h_7 &= h_2, \\ h_8 &= h_1, \\ h_9 &= h_0. \end{aligned}$$

The high-pass coefficients are

$$\begin{aligned} g_0 &= 0.0115088994 - 0.0193286077i + 0.0007856106j - 0.0052336021k, \\ g_1 &= -0.0391302581 - 0.0260294481i - 0.0039280528j + 0.0094926715k, \\ g_2 &= -0.0073656956 + 0.2133885981i + 0.0062848845j + 0.0081572006k, \\ g_3 &= 0.2393851082 - 0.1226724866i - 0.0166753392k, \\ g_4 &= -0.5109951348 - 0.3427619251i - 0.0109985479j - 0.0101062662k, \\ g_5 &= -g_4, \\ g_6 &= -g_3, \\ g_7 &= -g_2, \\ g_8 &= -g_1, \\ g_9 &= -g_0. \end{aligned}$$

51 These filters are used as quaternion-valued filters, not as four unrelated real filters. The left and
 52 right multiplication order is part of the definition of the transform.

53 **Nondecimated Quaternion Wavelet Transform.**

54 Let A_{j-1} denote the current quaternion image at level j . The level j dilated low-pass and
 55 high-pass filters induce nondecimated row-side convolution operators

$$\mathcal{H}_{1,j}, \quad \mathcal{G}_{1,j}, \quad (16)$$

56 and column-side convolution operators

$$\mathcal{H}_{2,j}, \quad \mathcal{G}_{2,j}. \quad (17)$$

57 For an $m \times n$ image, these operators preserve the original spatial dimensions. In the implementation
 58 the large convolution matrices are not formed explicitly. Their action is computed by circular shifts
 59 and quaternion multiplication by the filter coefficients.

60 The adjoint symbol $*$ denotes the conjugate-adjoint filtering operation. In the convolution
 61 implementation, the adjoint action reverses the circular shift indexing and conjugates the quaternion
 62 filter taps on the corresponding side.

63 At level j , the smooth and detail components are defined as

$$S_j = \mathcal{H}_{1,j} A_{j-1} \mathcal{H}_{2,j}^*, \quad (18)$$

$$H_j = \mathcal{G}_{1,j} A_{j-1} \mathcal{H}_{2,j}^*, \quad (19)$$

$$V_j = \mathcal{H}_{1,j} A_{j-1} \mathcal{G}_{2,j}^*, \quad (20)$$

$$D_j = \mathcal{G}_{1,j} A_{j-1} \mathcal{G}_{2,j}^*. \quad (21)$$

64 Here S_j is the smooth component. The component H_j is high-pass in the row direction and low-
 65 pass in the column direction. The component V_j is low-pass in the row direction and high-pass in
 66 the column direction. The component D_j is high-pass in both directions.

67 The approximation is updated recursively by

$$A_j = S_j. \quad (22)$$

68 Thus after L levels, the transform produces the final smooth component S_L and the detail compo-
 69 nents

$$\{H_j, V_j, D_j : j = 1, \dots, L\}. \quad (23)$$

70 Because the transform is nondecimated, each of these arrays has the same spatial dimension as the
 71 original image.

72 **Left and Right Quaternion Filtering.**

73 The noncommutativity of quaternions requires distinguishing row-side and column-side filtering.

74 If $f = \{f_r\}$ is a quaternion filter, then row-side filtering applies filter coefficients on the left:

$$(\mathcal{F}_{1,j}A)(x, y) = \sum_r f_r^{(j)} A(x - r, y). \quad (24)$$

75 The adjoint row-side action is

$$(\mathcal{F}_{1,j}^*A)(x, y) = \sum_r \bar{f}_r^{(j)} A(x + r, y). \quad (25)$$

76 Column-side filtering applies filter coefficients on the right:

$$(A\mathcal{F}_{2,j})(x, y) = \sum_r A(x, y - r) f_r^{(j)}. \quad (26)$$

77 The adjoint column-side action is

$$(A\mathcal{F}_{2,j}^*)(x, y) = \sum_r A(x, y + r) \bar{f}_r^{(j)}. \quad (27)$$

78 The exact signs in the shifts depend on the circular indexing used in the software implementation.
 79 What is essential is that the forward and inverse routines use the same indexing throughout. In
 80 the implementation used here, each two-dimensional component is obtained by applying the ap-
 81 propriate row-side operator and the appropriate adjoint column-side operator, as in the definitions
 82 of S_j, H_j, V_j , and D_j above [12, 2, 10, 9].

83 **Synthesis Formula.**

84 The inverse transform is implemented as the corresponding adjoint synthesis formula. For a
 85 tight or approximately tight nondecimated analysis pair, the level-wise reconstruction has the form

$$A_{j-1} = \frac{1}{4} \{ \mathcal{H}_{1,j}^* S_j \mathcal{H}_{2,j} + \mathcal{G}_{1,j}^* H_j \mathcal{H}_{2,j} + \mathcal{H}_{1,j}^* V_j \mathcal{G}_{2,j} + \mathcal{G}_{1,j}^* D_j \mathcal{G}_{2,j} \}. \quad (28)$$

86 The factor $1/4$ is the redundancy normalization associated with the two-dimensional nondecimated
 87 filtering step. The reconstruction is applied from the coarsest level back to the finest level. Starting
 88 from $A_L = S_L$, the algorithm successively reconstructs

$$A_{L-1}, A_{L-2}, \dots, A_0. \quad (29)$$

89 The recovered image A_0 is quaternion representation of the original image. In the numerical
 90 experiments, the reconstructed RGB image is obtained from the vector part of A_0 , namely the i ,
 91 j , and k components [15, 7, 17].

92 **Interpretation of Quaternion Wavelet Coefficients.**

93 Each wavelet coefficient is a quaternion

$$w = a + bi + cj + dk. \quad (30)$$

94 Its magnitude

$$|w| = \sqrt{a^2 + b^2 + c^2 + d^2} \quad (31)$$

95 measures the strength of the local multiscale response. The vector part

$$\text{Vec}(w) = bi + cj + dk \quad (32)$$

96 contains the coupled color-vector response. If $|\text{Vec}(w)| > 0$, the unit vector

$$u(w) = \frac{\text{Vec}(w)}{|\text{Vec}(w)|} \quad (33)$$

97 describes the local quaternion axis of the coefficient. These quantities motivate the feature sum-
98 maries described in the main text, including coefficient energy, magnitude distribution, phase con-
99 centration, axis summaries, directional balance, and orientation entropy.

100 The essential point is that the quaternion transform does not merely compute three separate
101 scalar wavelet transforms. It produces full quaternion coefficients whose scalar and vector compo-
102 nents arise from coupled color-spatial filtering. This is the main reason the quaternion nondecimated
103 representation is well matched to color histology images.

104 2 Mathematical Definitions of the Important QNDWT Features

105 This appendix gives the mathematical definitions of the QNDWT feature classes and individual
106 predictors that are used in the classification-results discussion. The goal is not to define every stored
107 feature in the feature table, but to define precisely the feature families and individual descriptors
108 that were retained by the radial-SVM permutation-importance summaries. In the group analysis,
109 the retained feature classes are those with mean decrease in held-out balanced accuracy above 5
110 percentage points. In the individual-predictor analysis, the retained predictors are those with mean
111 decrease in held-out balanced accuracy above 0.3 percentage points [6, 19, 3, 1].

112 Let A_0 be the pure quaternion image obtained from the RGB image. At decomposition level ℓ ,
113 let

$$Q_{H,\ell}(x, y), \quad Q_{V,\ell}(x, y), \quad Q_{D,\ell}(x, y) \quad (34)$$

114 denote the horizontal, vertical, and diagonal detail subbands, respectively. In the notation used
115 here, H is high-pass in the row direction and low-pass in the column direction, V is low-pass in
116 the row direction and high-pass in the column direction, and D is high-pass in both directions.
117 The smooth component at level ℓ is denoted by $S_\ell(x, y)$. Since the transform is nondecimated, all
118 subbands have the same pixel grid Ω as the original image. Let N_Ω denote the number of pixels in
119 this grid.

120 For any one coefficient in a subband, write

$$w = a + bi + cj + dk, \quad (35)$$

121 with scalar part a and vector part (b, c, d) . Define

$$\nu(w) = \sqrt{b^2 + c^2 + d^2}, \quad (36)$$

$$\rho(w) = |w| = \sqrt{a^2 + b^2 + c^2 + d^2}, \quad (37)$$

$$\theta(w) = \text{atan2}(\nu(w), a). \quad (38)$$

122 Here $\rho(w)$ is the quaternion amplitude and $\theta(w)$ is the scalar-vector phase angle. When $\nu(w) > 0$,
123 define the unit vector axis

$$u(w) = (u_i(w), u_j(w), u_k(w)) = \left(\frac{b}{\nu(w)}, \frac{c}{\nu(w)}, \frac{d}{\nu(w)} \right). \quad (39)$$

124 When $\nu(w) = 0$, the axis contribution is set to zero in the computation.

125 For compactness, if o denotes one of the three detail orientations H, V, D , write

$$w_{o,\ell}(x, y) = Q_{o,\ell}(x, y), \quad (40)$$

$$\rho_{o,\ell}(x, y) = \rho(w_{o,\ell}(x, y)), \quad (41)$$

$$\theta_{o,\ell}(x, y) = \theta(w_{o,\ell}(x, y)). \quad (42)$$

126 All averages below are over $(x, y) \in \Omega$ unless stated otherwise.

127 **2.1 Directional energy and orientation feature classes**

128 The detail energy in orientation o at level ℓ is

$$E_{o,\ell} = \frac{1}{N_\Omega} \sum_{(x,y) \in \Omega} \rho_{o,\ell}(x, y)^2. \quad (43)$$

129 The corresponding log-energy feature is

$$\log_2 E_{o,\ell} = \log_2 (\max(E_{o,\ell}, \epsilon_0)), \quad (44)$$

130 where ϵ_0 is a small positive numerical constant used only to avoid taking the logarithm of zero
131 [15, 7, 17].

132 At each level define the total detail energy

$$E_{T,\ell} = E_{H,\ell} + E_{V,\ell} + E_{D,\ell}. \quad (45)$$

133 The directional energy shares are

$$p_{H,\ell} = \frac{E_{H,\ell}}{E_{T,\ell} + \epsilon}, \quad (46)$$

$$p_{V,\ell} = \frac{E_{V,\ell}}{E_{T,\ell} + \epsilon}, \quad (47)$$

$$p_{D,\ell} = \frac{E_{D,\ell}}{E_{T,\ell} + \epsilon}. \quad (48)$$

134 The horizontal-vertical anisotropy is

$$A_{HV,\ell} = \frac{E_{H,\ell} - E_{V,\ell}}{E_{H,\ell} + E_{V,\ell} + \epsilon}. \quad (49)$$

135 The diagonal energy share is stored as

$$A_{D,\ell} = \frac{E_{D,\ell}}{E_{T,\ell} + \epsilon}. \quad (50)$$

136 Thus, in the current feature table, $A_{D,\ell}$ and $p_{D,\ell}$ have the same numerical definition, although
 137 they are retained under two mnemonic names. The orientation entropy at level ℓ is the normalized
 138 entropy of the three directional energy shares,

$$\text{OE}_\ell = -\frac{1}{\log 3} \sum_{o=H,V,D} p_{o,\ell} \log p_{o,\ell}, \quad (51)$$

139 with zero terms omitted from the sum [20].

140 The feature classes `Orientation_H_all`, `Orientation_V_all`, and `Orientation_D_all` collect all
 141 features attached to the corresponding detail orientation across levels, together with the associated
 142 global summaries for that orientation. Their high permutation importance means that the radial
 143 SVM is using not only the amount of detail energy, but also direction-specific amplitude, phase,
 144 and axis summaries. [6, 19, 3, 1]. The retained feature class `Orientation_anisotropy` consists of
 145 the directional shares $p_{H,\ell}, p_{V,\ell}, p_{D,\ell}$, the anisotropy index $A_{HV,\ell}$, the diagonal share $A_{D,\ell}$, and the
 146 orientation entropy OE_ℓ , together with their simple summaries across levels.

147 The retained level classes `Level_1_all`, `Level_2_all`, `Level_5_all`, and `Level_6_all` contain all
 148 QNDWT features at the corresponding decomposition level. Thus a level class combines the three
 149 detail orientations, their amplitude summaries, phase summaries, axis summaries, and orientation-
 150 balance summaries at one spatial scale. Levels 1 and 2 represent relatively fine detail. Levels 5 and
 151 6 represent coarser tissue organization.

152 2.2 Amplitude distribution features

153 For a fixed orientation o and level ℓ , the amplitude sample is

$$\rho_{o,\ell}(x, y), \quad (x, y) \in \Omega. \quad (52)$$

154 The amplitude mean and standard deviation are

$$\bar{\rho}_{o,\ell} = \frac{1}{N_\Omega} \sum_{(x,y) \in \Omega} \rho_{o,\ell}(x,y), \quad (53)$$

$$s_{o,\ell} = \left(\frac{1}{N_\Omega} \sum_{(x,y) \in \Omega} (\rho_{o,\ell}(x,y) - \bar{\rho}_{o,\ell})^2 \right)^{1/2}. \quad (54)$$

155 The retained feature class **Amplitude_std** consists of $s_{o,\ell}$ over orientations $o = H, V, D$ and levels
 156 $\ell = 1, \dots, L$. This class measures spatial variability of QNDWT response magnitudes within each
 157 subband.

158 The sample skewness and kurtosis of amplitudes are

$$\gamma_{1,o,\ell} = \frac{1}{N_\Omega} \sum_{(x,y) \in \Omega} \left(\frac{\rho_{o,\ell}(x,y) - \bar{\rho}_{o,\ell}}{s_{o,\ell}} \right)^3, \quad (55)$$

$$\beta_{2,o,\ell} = \frac{1}{N_\Omega} \sum_{(x,y) \in \Omega} \left(\frac{\rho_{o,\ell}(x,y) - \bar{\rho}_{o,\ell}}{s_{o,\ell}} \right)^4. \quad (56)$$

159 When the amplitude variance is zero, these quantities are set to zero in the implementation. The
 160 individual predictors **ampKurt_H_L2** and **ampSkew_H_L5** are examples of these two definitions. The
 161 first is the amplitude kurtosis in the H subband at level 2, and the second is the amplitude skewness
 162 in the H subband at level 5.

163 The amplitude entropy is computed from a fixed 64-bin histogram of the amplitude sample in
 164 the subband. If π_m is the empirical proportion in bin m , then

$$H_{\rho,o,\ell} = -\frac{1}{\log 64} \sum_{m=1}^{64} \pi_m \log \pi_m, \quad (57)$$

165 again omitting empty bins from the sum. The individual predictor **ampEntropy_H_L4** is $H_{\rho,H,4}$, the
 166 normalized histogram entropy of horizontal-detail amplitudes at level 4.

167 2.3 Quaternion phase concentration and phase variance

168 The phase feature class retained by group permutation importance is **Phase_concentration_variance**.
 169 For a fixed subband $Q_{o,\ell}$, the circular phase summaries are computed with amplitude weights. De-
 170 fine

$$C_{o,\ell} = \frac{\sum_{(x,y) \in \Omega} \rho_{o,\ell}(x,y) \cos \theta_{o,\ell}(x,y)}{\sum_{(x,y) \in \Omega} \rho_{o,\ell}(x,y) + \epsilon}, \quad (58)$$

$$S_{o,\ell} = \frac{\sum_{(x,y) \in \Omega} \rho_{o,\ell}(x,y) \sin \theta_{o,\ell}(x,y)}{\sum_{(x,y) \in \Omega} \rho_{o,\ell}(x,y) + \epsilon}. \quad (59)$$

171 The amplitude-weighted mean phase is

$$\bar{\theta}_{o,\ell} = \text{atan2}(S_{o,\ell}, C_{o,\ell}), \quad (60)$$

172 although the mean phase itself was not among the retained top feature classes. The phase concen-
173 tration is

$$R_{o,\ell} = \sqrt{C_{o,\ell}^2 + S_{o,\ell}^2}, \quad (61)$$

174 and the circular phase variance is

$$V_{\theta,o,\ell} = 1 - R_{o,\ell}. \quad (62)$$

175 The individual predictors `phaseConc_DL1` and `phaseVar_DL1` are $R_{D,1}$ and $V_{\theta,D,1}$, respectively.
176 They measure, in complementary form, how coherent the scalar-vector phase angles are in the
177 diagonal detail band at the finest scale.

178 2.4 Quaternion axis features

179 For a fixed subband, the unit axis $u(w)$ records the direction of the vector part of the quaternion
180 coefficient. The amplitude-weighted mean axis is

$$\bar{u}_{o,\ell} = (\bar{u}_{i,o,\ell}, \bar{u}_{j,o,\ell}, \bar{u}_{k,o,\ell}) \quad (63)$$

$$= \frac{\sum_{(x,y) \in \Omega} \rho_{o,\ell}(x,y) (u_i(w_{o,\ell}(x,y)), u_j(w_{o,\ell}(x,y)), u_k(w_{o,\ell}(x,y)))}{\sum_{(x,y) \in \Omega} \rho_{o,\ell}(x,y) + \epsilon}. \quad (64)$$

181 The corresponding axis concentration is

$$R_{u,o,\ell} = (\bar{u}_{i,o,\ell}^2 + \bar{u}_{j,o,\ell}^2 + \bar{u}_{k,o,\ell}^2)^{1/2}. \quad (65)$$

182 Because the RGB image is coded as $Ri + Gj + Bk$, the i , j , and k components of the axis are
183 aligned with the red, green, and blue quaternion directions, but they are not simple channel means.
184 They are amplitude-weighted summaries of the direction of the vector part after coupled quaternion
185 filtering.

186 Several retained individual predictors are axis components. Specifically, `axisMean_j_V_L2` is
187 $\bar{u}_{j,V,2}$, `axisMean_k_H_L1` is $\bar{u}_{k,H,1}$, and `axisMean_j_V_L3` is $\bar{u}_{j,V,3}$. These predictors describe whether
188 the vector parts of the important QNDWT coefficients point preferentially in a particular color-
189 quaternion direction within a particular orientation and scale.

190 2.5 RGB marginal features

191 The retained individual RGB predictors are computed directly from the normalized input image,
192 before QNDWT filtering. If $R(x,y)$, $G(x,y)$, and $B(x,y)$ denote normalized red, green, and blue

193 intensities, then

$$\bar{R} = \frac{1}{N_\Omega} \sum_{(x,y) \in \Omega} R(x,y), \quad (66)$$

$$\bar{G} = \frac{1}{N_\Omega} \sum_{(x,y) \in \Omega} G(x,y), \quad (67)$$

$$\bar{B} = \frac{1}{N_\Omega} \sum_{(x,y) \in \Omega} B(x,y). \quad (68)$$

194 The channel standard deviations are

$$s_R = \left(\frac{1}{N_\Omega} \sum_{(x,y) \in \Omega} (R(x,y) - \bar{R})^2 \right)^{1/2}, \quad (69)$$

$$s_G = \left(\frac{1}{N_\Omega} \sum_{(x,y) \in \Omega} (G(x,y) - \bar{G})^2 \right)^{1/2}, \quad (70)$$

$$s_B = \left(\frac{1}{N_\Omega} \sum_{(x,y) \in \Omega} (B(x,y) - \bar{B})^2 \right)^{1/2}. \quad (71)$$

195 The red-green correlation is

$$r_{RG} = \frac{\sum_{(x,y) \in \Omega} (R(x,y) - \bar{R}) (G(x,y) - \bar{G})}{\left[\sum_{(x,y) \in \Omega} (R(x,y) - \bar{R})^2 \sum_{(x,y) \in \Omega} (G(x,y) - \bar{G})^2 \right]^{1/2}}. \quad (72)$$

196 The individual predictors `mean_R`, `mean_B`, `std_G`, `std_B`, and `corr_RG` correspond to $\bar{R}$, $\bar{B}$, s_G , s_B ,
 197 and r_{RG} , respectively. Their presence among the retained individual predictors indicates that global
 198 stain and color-balance information remains useful even when the classifier also uses multiscale
 199 quaternion texture descriptors.

200 2.6 Across-level directional summaries

201 The important directional feature classes also include global summaries across decomposition levels.
 202 For orientation o , define the across-level energy vector

$$E_o = (E_{o,1}, E_{o,2}, \dots, E_{o,L}). \quad (73)$$

203 The normalized spectral entropy of this vector is

$$\text{SE}_o = -\frac{1}{\log L} \sum_{\ell=1}^L q_{o,\ell} \log q_{o,\ell}, \quad (74)$$

$$q_{o,\ell} = \frac{E_{o,\ell}}{\sum_{r=1}^L E_{o,r} + \epsilon}. \quad (75)$$

204 The across-level log-energy slope is the fitted least-squares slope in the regression

$$\log_2 E_{o,\ell} = \alpha_o + b_o \ell + e_{o,\ell}, \quad \ell = 1, \dots, L. \quad (76)$$

205 The stored slope feature is b_o . For the total detail energy $E_{T,\ell}$, the same definitions give SE_T and
206 b_T . The smooth energy after the final decomposition level is

$$E_{S,L} = \frac{1}{N_\Omega} \sum_{(x,y) \in \Omega} |S_L(x,y)|^2, \quad (77)$$

207 with log version $\log_2 E_{S,L}$ defined in the same way as the detail log-energy features. These across-
208 level and smooth-energy summaries are part of the global directional summaries used by the retained
209 orientation feature classes [15, 7, 17, 21].

210 2.7 Summary of retained individual predictors

211 The individual predictors exceeding the 0.3 percentage-point threshold in the radial-SVM permuta-
212 tion analysis were screened for exact functional redundancy. After removing functionally dependent
213 summaries, the retained individual predictors are

$$\begin{aligned} \text{axisMean_j_V_L2} &= \bar{u}_{j,V,2}, \\ \text{axisMean_k_H_L1} &= \bar{u}_{k,H,1}, \\ \text{axisMean_j_V_L3} &= \bar{u}_{j,V,3}, \\ \text{std_G} &= s_G, \\ \text{mean_R} &= \bar{R}, \\ \text{std_B} &= s_B, \\ \text{ampEntropy_H_L4} &= H_{\rho,H,4}, \\ \text{A_D_L5} &= \frac{E_{D,5}}{E_{T,5} + \epsilon}, \\ \text{mean_B} &= \bar{B}, \\ \text{phaseVar_D_L1} &= 1 - R_{D,1}, \\ \text{ampKurt_H_L2} &= \beta_{2,H,2}, \\ \text{corr_RG} &= r_{RG}, \\ \text{ampSkew_H_L5} &= \gamma_{1,H,5}. \end{aligned}$$

214 Two summaries that also exceeded the 0.3 percentage-point threshold were not retained in this
215 reduced list because they are functionally dependent on predictors already listed: `p_D_L5` is identical
216 to `A_D_L5` under the present feature definitions, and `phaseConc_D_L1` = 1 − `phaseVar_D_L1`.

217 The retained set is therefore a mixture of quaternion-axis summaries, RGB stain summaries,
218 orientation-energy summaries, diagonal phase variability, and amplitude-shape summaries. This is
219 consistent with the interpretation in the main results section that the radial SVM uses both color
220 information and multiscale quaternion geometry.

- **axisMean_j_V_L2**. The feature **axisMean_j_V_L2** is the amplitude-weighted mean of the j component of the normalized quaternion vector axis, computed from the level 2 vertical detail subband. In the quaternion image coding, the RGB image is represented as the pure quaternion field

$$A_0(x, y) = R(x, y)i + G(x, y)j + B(x, y)k,$$

so the j direction initially corresponds to the green channel. The symbol V indicates that the feature is computed from the vertical-oriented QNDWT detail coefficients, and **L2** indicates decomposition level 2.

This feature does not represent simply the average green intensity. Rather, it measures how strongly the green color direction participates in level 2 vertical multiscale texture. In histopathology images, this can be discriminatory because diagnostic classes may differ in the color balance of oriented local structures, such as nuclei, gland boundaries, epithelial texture, and stromal patterns. Thus, **axisMean_j_V_L2** can capture a class-dependent green-axis contribution to vertical tissue microstructure, rather than a simple marginal color effect. [2, 9, 10, 11, 23].

Analogous description for the **axisMean_k_H_L1** and **axisMean_j_V_L3**.

- **std_G**. The feature **std_G** is the sample standard deviation of the normalized green-channel intensities over all pixels in the image. If G_r denotes the green intensity at pixel r and $\bar{G}$ is the image-wide green-channel mean, then

$$s_G = \left\{ \frac{1}{N-1} \sum_{r=1}^N (G_r - \bar{G})^2 \right\}^{1/2}.$$

Thus, **std_G** measures the amount of green-channel variability across the tissue image. It may be histopathologically significant because it reflects spatial heterogeneity in staining, tissue density, nuclear-cytoplasmic contrast, stromal variation, and local textural irregularity. In histopathology images, diagnostically different tissue classes may have similar average color levels but different degrees of color dispersion; therefore, green-channel variability can contribute useful discriminatory information beyond the mean intensity alone.

The interpretation of **mean_R**, **mean_B**, and **std_B** is analogous. The mean features summarize the average normalized red and blue intensities in the image, while **std_B** summarizes the image-wide variability of the blue channel. Their diagnostic relevance comes from the fact that red and blue channel levels and dispersions may reflect different aspects of stain uptake, nuclear prominence, cytoplasmic or stromal content, and tissue heterogeneity. [18, 11, 23, 14]

mean_R, **mean_G**, **mean_B**, **std_R**, **std_G**, and **std_B** are not wavelet features. They are measured directly on the original normalized RGB image, before applying the quaternion nondecimated wavelet transform. They are global marginal color summaries over all image pixels.

- **ampEntropy_H_L4**. The feature **ampEntropy_H_L4** is the normalized entropy of the empirical distribution of quaternion coefficient amplitudes in the horizontal detail subband at decomposition

level 4. For each coefficient in this subband, the quaternion amplitude is

$$M_{H,4,r} = |w_{H,4,r}|,$$

where $r = 1, \dots, N$ indexes all image pixels. The values $M_{H,4,r}$ are binned into a 64-bin histogram, and the normalized entropy is computed from the resulting bin probabilities.

This feature measures how dispersed or heterogeneous the amplitude values are in the level 4 horizontal detail component. A small value indicates that the horizontal detail amplitudes are concentrated in a narrow range, while a large value indicates a more diverse distribution of horizontal multiscale texture strengths. In histopathology images, `ampEntropy_H_L4` may be discriminatory because different diagnostic classes can exhibit different degrees of organized or irregular horizontal tissue texture at an intermediate scale [18, 11, 23].

- **A_D_L5**. The feature `A_D_L5` is the diagonal-detail energy share at decomposition level 5. If $E_{H,5}$, $E_{V,5}$, and $E_{D,5}$ denote the horizontal, vertical, and diagonal detail energies at level 5, then

$$T_5 = E_{H,5} + E_{V,5} + E_{D,5},$$

and

$$A_{D,5} = \frac{E_{D,5}}{T_5 + \varepsilon}.$$

Thus, `A_D_L5` measures the proportion of level 5 detail energy that is carried by the diagonal subband. In the present implementation, this quantity is numerically the same as `p_D_L5`.

This feature represents the relative importance of diagonal multiscale texture at a relatively coarse level of the QNDWT decomposition. A larger value indicates that the level 5 tissue structure contains comparatively stronger diagonal detail, while a smaller value indicates that the detail energy is concentrated more in the horizontal or vertical subbands. In histopathology images, `A_D_L5` may be discriminatory because different diagnostic classes can differ in the organization, orientation, and irregularity of glands, epithelial boundaries, stromal patterns, and nuclear arrangements at coarser spatial scales. [15, 7, 17, 2, 10, 11, 23]

- **phaseVar_D_L1**. The feature `phaseVar_D_L1` is the circular variance of the quaternion phase angles in the diagonal detail subband at decomposition level 1. For a quaternion coefficient

$$w_{D,1,r} = a_{D,1,r} + b_{D,1,r}i + c_{D,1,r}j + d_{D,1,r}k,$$

the phase angle is defined by

$$\theta_{D,1,r} = \text{atan2}(\rho_{D,1,r}, a_{D,1,r}),$$

where

$$\rho_{D,1,r} = (b_{D,1,r}^2 + c_{D,1,r}^2 + d_{D,1,r}^2)^{1/2}.$$

The phase concentration is computed from the amplitude-weighted circular average of these phase angles, and `phaseVar_D_L1` is

$$V_{\theta,D,1} = 1 - R_{\theta,D,1}.$$

Thus, larger values of **phaseVar_D_L1** indicate more dispersed quaternion phase behavior in the level 1 diagonal detail coefficients, while smaller values indicate more coherent phase behavior.

This feature represents the variability of scalar-vector phase structure in the finest-scale diagonal texture component. Since level 1 captures fine local detail, and the diagonal subband captures diagonal edge and texture behavior, **phaseVar_D_L1** may be discriminatory when diagnostic classes differ in fine-scale diagonal tissue irregularity, nuclear texture, glandular boundaries, or stain-transition patterns. It is therefore not simply a color feature or an energy feature, but a measure of how consistently the quaternion coefficient phases behave within fine diagonal histopathological texture [8, 16, 2, 10].

- **ampKurt_H_L2**. The feature **ampKurt_H_L2** is the kurtosis of the quaternion coefficient amplitudes in the horizontal detail subband at decomposition level 2. For each coefficient in this subband, let

$$M_{H,2,r} = |w_{H,2,r}|, \quad r = 1, \dots, N,$$

denote its quaternion amplitude. If $\bar{M}_{H,2}$ is the mean amplitude and

$$\mu_{k,H,2} = \frac{1}{N} \sum_{r=1}^N (M_{H,2,r} - \bar{M}_{H,2})^k$$

is the k th central moment, then

$$\text{ampKurt_H_L2} = \frac{\mu_{4,H,2}}{\mu_{2,H,2}^2}.$$

This feature measures the tail-heaviness or peakedness of the amplitude distribution in the level 2 horizontal detail component. A large value indicates that many horizontal detail amplitudes are relatively small, but a few are very large, suggesting sparse strong horizontal edges or localized texture contrasts. A smaller value indicates a more regular and less extreme distribution of horizontal detail amplitudes. In histopathology images, **ampKurt_H_L2** may be discriminatory because diagnostic classes can differ in the presence of sharp horizontal gland boundaries, elongated epithelial structures, stromal bands, nuclear alignments, or other localized horizontal tissue patterns at a fine to intermediate scale [13, 22].

- **corr_RG**. The feature **corr_RG** is the Pearson correlation between the normalized red-channel and green-channel pixel intensities in the original RGB image. If R_r and G_r denote the red and green intensities at pixel r , for $r = 1, \dots, N$, then

$$\text{corr_RG} = \frac{\sum_{r=1}^N (R_r - \bar{R})(G_r - \bar{G})}{\left\{ \sum_{r=1}^N (R_r - \bar{R})^2 \sum_{r=1}^N (G_r - \bar{G})^2 \right\}^{1/2}}.$$

Here $N = mn$ is the total number of pixels in the image. This is a global marginal color feature computed before the QNDWT decomposition, not a wavelet-level feature.

The feature measures how strongly red and green intensities co-vary across the tissue image. A large positive value indicates that pixels with high red intensity tend also to have high green

intensity, while a smaller value indicates weaker channel co-movement. In histopathology images, **corr_RG** may be discriminatory because diagnostic classes can differ in stain balance, nuclear-cytoplasmic contrast, stromal composition, and tissue heterogeneity. Thus, it captures joint red-green color organization rather than the behavior of either channel alone. [18, 11, 23, 14, 18].

- **ampSkew_H.L5**. The feature **ampSkew_H.L5** is the skewness of the quaternion coefficient amplitudes in the horizontal detail subband at decomposition level 5. For each coefficient in this subband, let

$$M_{H,5,r} = |w_{H,5,r}|, \quad r = 1, \dots, N,$$

denote its quaternion amplitude. If $\bar{M}_{H,5}$ is the mean amplitude and

$$\mu_{k,H,5} = \frac{1}{N} \sum_{r=1}^N (M_{H,5,r} - \bar{M}_{H,5})^k$$

is the k th central moment, then

$$\text{ampSkew_H.L5} = \frac{\mu_{3,H,5}}{\mu_{2,H,5}^{3/2}}.$$

This feature measures the asymmetry of the amplitude distribution in the level 5 horizontal detail component. A positive value indicates a right-skewed distribution, meaning that most horizontal detail amplitudes are relatively small, but some coefficients are much larger. This may correspond to sparse but strong horizontal tissue structures at a relatively coarse scale. In histopathology images, **ampSkew_H.L5** may be discriminatory because diagnostic classes can differ in the presence and spatial organization of coarse horizontal gland boundaries, stromal bands, epithelial arrangements, or elongated tissue structures [13, 22, 11, 23].

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